Manuscript outline: what each paragraph should contain

enceladus_DSM `studies/keff_sintering/manuscript_outline.tex`

2026-10-13, against the Overleaf project uploaded that day

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,^{\circ}\mathrm{C}$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. Every paragraph of the draft has a numbered entry: its subject, where it is, a one-line overview, and bullets for what I suggest changing. Line numbers (L first–last) are those of the source of 2026-10-13 compiled here (32 pages; the body ends at L508). Each bullet is tagged: **[correct]** a statement that is wrong or unsupported; **[wording]** grammar or wording; **[trim]** text that can go or shrink; **[add]** something missing. A paragraph with no **[correct]** bullet says nothing I found to be wrong. Entries marked "to write" do not exist yet; their bullets are the content. Boxes mark where a figure or table goes. From Section 5.2, paragraph 5, on, the entries keep the earlier form (a "Message" and a content list), since nothing is drafted there.

The old Discussion is gone from the source. Its literature points are kept here, marked **[from the notes]** . Their line numbers refer to the PDF of 2026-10-12 in `Literature/`, which still contains that text. Each point is to be checked against its source before use.

Suggested citations follow each paragraph, with the key in your `.bib`. Most are the ones the old Discussion used. I have matched them by title only; check each claim against the paper before it goes in.

Decisions already made, which this plan follows

Collected from the work of 2026-09-08 to 2026-10-12: the session log, the campaign record and the manuscript plan.

- **Framing** (10-08, 10-10). Laboratory-observable sintering by vapor transport, inside the dry-snow-metamorphism story. The Introduction and title stay as written; scope caveats go in the Methods. No timescales for Enceladus temperatures; the timescale map is dropped. Advisor's comments still to come.
- **Comparisons** (10-08). Not natural snow (angular, more porous, loaded). The analogues are unloaded ice-sphere and ice-powder samples: Molaro et al., Demmenie et al., Choukroun et al. From the snow literature only functional forms carry over, never magnitudes.
- **Kinetics** (09-22, 10-09). α_c constant in temperature; the thin-interface term always on; aggregates stated as $\alpha_c = 1.020 \times 10^{-3}$. No further simulations for this paper.
- **Two-grain argument** (09-09, 10-02, 10-13). Rests on what the vapor route does and does not reproduce, and on exponents; no Thomas or Kingery runs. The two mismatches have different

causes and neither is attributed to missing surface diffusion (Appendix, this outline): the rate at $-20\,^{\circ}\mathrm{C}$ is a matter of how fast vapor can be supplied, in an experiment whose driving force is a few millikelvin-equivalent; the exponent at saturation is, on present evidence, an artifact of starting from a pre-made neck (to be confirmed by the tangent-start run). Molaro's conventions: neck *width*, time anchored on the first measurement. The $-20\,^{\circ}\mathrm{C}$ series is the benchmark.

- **Conductivity** (09-23). Tensor law in the interface; the cell problem is a steady problem on a frozen microstructure and says nothing about the transient model. Two dimensions only, so no absolute k_{eff} is claimed.
- **Clock and normalization** (10-08, 10-12). The clock is τ_{sub}; curves are normalized at $\theta_r = 30$. The first day depends on the interface parameter, so a "rise from day 1" is never quoted as a property of the ice.
- **Notation** (09-28 to 09-30). ϕ phase field, φ porosity, ζ indicator, χ curvature; subscripts "ice" and "air", not i and a; no index notation; a single dot for every contraction; the vector $\mathbf{t}$ named as in Auriault et al.
- **Citations** (09-23). Only work whose setting matches ours; every DOI checked against Crossref.
- **Words.** Desublimation ("deposition" is plume fallout); "vapor"; ε is the interface parameter, not the interface width.

Where the draft stands

Section	State
Key points, Abstract, Summary	to write (placeholders)
1 Introduction	written; small fixes and trims remain
2.1–2.3 Model, kinetics, homogenization	written; four definitions to add
2.4, 2.5 Numerics, simulation design	headings only
3 Grain pairs	written; numbers to add, one sentence wrong
4 Aggregates	written; numbers to add, one paragraph to correct
5.1 What vapor transport reproduces	three paragraphs drafted; the first two need revising for what was found on 10-13
5.2 Temperature as a rescaling of time	one paragraph begun
5.3–5.6, 6 Conclusion	headings only
Appendices A, B	headings only; content is in the Appendix of this outline

Done since the last check: the section structure of this outline is adopted; the line-breaking fix and `width=\textwidth` figures are in, and the project compiles with no overfull boxes, no undefined citations and no undefined references; the temperature table is placed; the Figure 3 caption covers panels (c), (d); Figure 1 says $w(t)$; "30-day"; "harbor".

Work list, in order

1. **Correct Section 5.1 as drafted** (L442–493). Two sentences attribute the shortfall to surface diffusion, which Appendix A rules out; "by 41% and 170%" misstates the numbers; one sentence is unfinished. Details under Section 5.1 below.
2. **Correct Section 3.2** (L346–347): $\alpha_c = 10^{-3}$ is the attachment-limited regime, not a vapor-limited one.
3. **Wait for the tangent-start saturated run** before finalizing what Sections 3.2, 5.1 and Appendix B say about the exponent ("Runs pending" below).
4. **Write Sections 2.4 and 2.5**; move the set-up text out of Sections 3 and 4 into them.
5. **Add the numbers to Section 4** and correct the high-porosity paragraph (L398–416).
6. **Finish 5.2; write 5.3–5.6, the Conclusion, Appendices A and B.** Plans below; the two-page scaling note (`temperature_scaling/`) is the source for 5.2.
7. **Front matter:** key points, abstract, plain-language summary.
8. **Small corrections** (last section of this outline).

Runs pending

- **Saturated pair from tangent contact** (submitted 2026-10-13; batch file `demmenie_tangent_T-20`). Same grains, interface parameter and kinetics as the run in Figure 4; the grains start just touching. It decides between two readings of the low exponent: the pre-made neck (its shape, and the unknown time it "had been growing") or the interface width. Test: at equal neck widths, compare each run's growth rate with the Gibbs–Thomson value. Expected exponent about 0.28.
- **Molaro $-20\,^\circ$C from tangent contact** (prepared, not submitted). A 10-minute pilot (batch file `molaro_tangent_T-20_pilot`) checks first for a trapped void at the neck, the artifact that led to the pre-made neck in August.
- Until the first is in, write the exponent passages as marked "provisional" below.

Section titles

The descriptive headings are adopted in the draft. Still to change: "2 Methods" (suggested: Modeling Approach), "2.2 Gibbs-Thomson Condition" (Interface Kinetics and the Sintering Time Scale), "2.3 Homogenizing for Effective Thermal Conductivity" (Effective Thermal Conductivity by Homogenization), and capitals in "5.1 What vapor transport alone reproduces". All optional.

Build check (LaTeX warnings)

The project uploaded on 2026-10-13 compiles here in 32 pages with **no overfull or underfull boxes, no undefined citations, no undefined references**, and BibTeX reports no warnings. What remains:

1. **Two font warnings** ("Font shape … in size ⟨10.5⟩ not available"). The class sets section headings at 10.5 pt, a size Computer Modern does not have, so LaTeX substitutes 10.95 pt. Harmless. To remove them, put this on the line *before* `\documentclass` (tested: zero warnings):

   ```
   \RequirePackage{fix-cm}
   ```

2. **"Label(s) may have changed. Rerun"** appears after any edit that moves a label and goes away on the next compile. Overleaf reruns automatically; if it persists there, recompile from scratch (the "Recompile" menu, "Clear cached files").
3. **Not cited in the text** (no warning, but AGU requires it): Table 2 (temperature-dependent quantities), Table 3 (complete parameters), Figure 2.
4. **Unused files** in the project: the Figure 8 timescale map, the all-porosity variant of Figure 7, and the AGU template file.

If Overleaf shows warnings other than these, paste them: its compiler settings can differ from the build here.

Front matter

1. **Title and authors** (L1–4)

   The paper's name and author list.

   - [wording] "Modeling The Impact": lower-case "the". Remove "–Version 2".
   - [wording] "Ahmed, Mahjoub": stray comma inside one author's name.

2. **Key points** (L7–8, placeholder)

   Three statements of at most 140 characters each.

   - [add] For a given ice-grain packing, temperature from -40 to $-5\,^{\circ}\mathrm{C}$ only rescales the time over which sintering raises conductivity.
   - [add] Effective conductivity follows specific surface area by one power law in well-connected packings; porosity sets its level.
   - [add] Vapor transport alone reproduces part of the neck growth measured between ice grains in the laboratory.

3. **Abstract** (L9–31, marked "needs to be updated")

   Background, method, the two-grain test, the aggregate result.

   - [correct] "Driven by temperature gradients": every run here is isothermal.
   - [correct] "replicating experimental results for various temperatures without retuning" and "incorporating a temperature dependence into our kinetic parameters measured in experiments": neither is done; α_c is constant and the model gives 41% of the measured growth.
   - [correct] "collapse … for both varying temperatures and varying porosities": across temperature yes; across porosity only the exponent is shared, for $\varphi \leq 0.375$.
   - [trim] The first five sentences are background; keep two.

- **[add]** Sentence by sentence: setting; gap (conductivity as grains sinter without compaction); method; grain pairs (41% at $-20\,^{\circ}\mathrm{C}$; exponent at saturation); aggregates (125 runs); result 1 (same conductivity against SSA at every temperature within 0.3%); result 2 ($\mathrm{SSA}^{-0.80}$ for $\varphi \leq 0.375$, level set by porosity); meaning. At most 250 words (it is 254 now).

4. **Plain Language Summary** (L32–55, marked "needs to be updated")

   The same story for a general reader.

   - **[correct]** "tune our model at $-20\,^{\circ}\mathrm{C}$ … without retuning … strong agreement" and "regardless of temperature or initial porosity": as for the abstract.
   - **[correct]** "changes in temperatures (sunrise or sunset)" and "gradients in temperature": not part of this study.
   - **[wording]** "know as" (known); "effecting", "effects" twice, "effect" (affecting, affects, affect); "C.Then" (space).
   - **[trim]** About 300 words; the limit is 200.

1 Introduction

Kept as drafted in structure; no reordering.

1. **What dry snow metamorphism is** (L57–68)

   Defines the process, gives everyday examples, and says it changes bulk properties.

   - **[correct]** "thermophysical properties (e.g., porosity, …)": porosity does not change by this process without compaction, and it is fixed in every run here. Drop it from the list.
   - **[correct]** "hoar frost resulting in slab avalanches": the weak layer is depth hoar (as paragraph 2 says).
   - **[wording]** "evolves the thermophysical properties": "changes". "interfacial surface area": "interfacial area".
   - **[trim]** The ice-cream, snow and avalanche examples return in paragraph 2; keep them in one place.

2. **Why it matters on Earth** (L69–81)

   Avalanches, firn, and heat flow through snowpacks.

   - **[wording]** "In addition to growing frost on your ice cream": informal for JGR. "therefore is important": "is therefore important". "viability of ice": "persistence of ice".
   - **[trim]** The firn-aquifer sentence concerns liquid water, which is outside dry metamorphism; it can go.

3. **Icy bodies and Enceladus** (L82–101)

   Water ice across the solar system, the plume, and where sintering is expected.

   - **[wording]** "However, near the Tiger Stripes, where temperatures may reach up to 180 K, can result in significant sintering" has no subject: "temperatures near the Tiger Stripes reach about 180 K, enough for significant sintering".
   - **[wording]** "where the mission had concluded numerous jets converging": "which observed numerous jets merging". "coined": "known as". "contains": "contain". "Ocean bearing", "micron sized": hyphenate. "effects": "affects".

- [trim] "giving rise to the importance for understanding icy systems on planetary bodies": cut. The last sentence is circular (understanding bulk properties aids understanding bulk properties): cut.

4. **Sintering as part of metamorphism** (L102–116)

   Neck growth between grains, its two routes, and earlier work.

   - [correct] "The dominating modes . . . vapor diffusion . . . and surface diffusion": by Molaro's own equations surface diffusion dominates only while necks are a few micrometres or less (Appendix A). Say "vapor transport, and surface diffusion while necks are small".
   - [correct] "Other modes of diffusion tend to have limited effects or operate over planetary timescales": needs a citation (Molaro et al.).
   - [correct] "primarily focused on terrestrial settings": the list includes Molaro and Demmenie, which are laboratory studies of grain pairs.
   - [wording] "a pair ice grains" (pair of); "deposits on the neck" (desublimates); "has been the study of much research" (has been studied extensively).
   - [trim] The last sentence restates paragraph 1.
   - [add] One clause: Demmenie et al. is the second comparison used here.

   Figure 1 [written] placed. [wording] Caption: "A 2-D cross" (cross-section); "neck of radius $w(t)$" (w is the width).

5. **Attachment kinetics** (L117–134)

   Supersaturation, the attachment law, and the condensation coefficient.

   - [correct] "Wilson-Frenkel relation": check the name. For growth from vapor, $v_n = \alpha_c v_{\mathrm{kin}} \sigma$ is usually called the Hertz–Knudsen relation (Libbrecht, 2017; Molaro's Appendix A); Wilson–Frenkel is the analogous law for growth from the melt.
   - [correct] "suppresses the attachment kinetics well below the vapor transport limited regime": it means growth is limited by attachment, not by transport.
   - [correct] "Libbrecht (2017) led an experiment in near vacuum": that reference is a review; the measurements are Libbrecht & Rickerby (2013), which is in the `.bib`.
   - [correct] The paragraph builds toward a temperature-dependent α_c through $\sigma_0(T)$. The model holds α_c constant. End on that: α_c is uncertain over two decades (10^{-3}–10^{-1}) and is held constant within each study here.
   - [wording] "Vapor molecules do not always nucleate on icy surfaces due to their hexagonal crystallographic structure": say "on faceted surfaces a new molecular layer must nucleate before it can grow". "super saturation" (one word). "temperature dependence between" (of).
   - [trim] Eighteen lines; six carry what the paper uses.

6. **Metamorphism beyond a pair; earlier models** (L135–154)

   From a grain pair to an aggregate, and the models that exist.

   - [correct] "densification, especially in the presence of overburden pressures or temperature gradients": a temperature gradient recrystallizes snow at constant density; drop "or temperature gradients".

- [wording] "in attempt to" twice (to); "it was required to tune" (both tuned); `\cite` where the rest use `\citet`.
- [trim] The first two sentences repeat paragraph 4. "This is a very rich problem ..." adds nothing.

7. **What is not known** (L155–166)

   The gap the paper addresses.

   - [correct] "It is still unknown how metamorphism influences the macroscopic properties in the absence of compaction": too strong; Calonne et al. (2014) followed conductivity at constant density under a gradient. Say it is not quantified for isothermal sintering.
   - [correct] "no model can accurately predict ... for varying temperatures and varying vapor saturation": this paper does not close that gap; do not open it.
   - [correct] Bouvet is cited as 2023 here and 2022 in paragraph 6 for the same statement.
   - [wording] "effects" (affects); "macroscopic bodies of icy porous materials" (bulk properties); "here on earth" (Earth); "As of current" (At present); "Significant work ... are needed" (is).
   - [trim] The sentence on Bouvet and Braun tuning α_c repeats paragraph 6.

8. **What follows** (L167–180)

   The order of the paper.

   - [correct] "is dependent of the mean temperature field or the initial porosity": independent of temperature; of porosity only in its exponent.
   - [correct] "use their results to validate model": "compare the model with".
   - [wording] Delete the fragment that ends the paragraph ("d theoretically, are needed ..."). "over 2D packings ... over 30-day time periods". Four sentences begin "We then".
   - [add] The saturated grain pair is not announced.

2 Modeling Approach

Draft heading: 2 Methods.

1. **Section opening** (L182–189)

   Says the model follows Kaempfer & Plapp and Moure & Fu.

   - [wording] "The scientific novelty in this work is much more an application of the model, rather than advancing the numerical methods": "The contribution here is the application of the model, not the method".
   - [trim] The last three sentences are repeated in the first lines of Section 2.1. Three sentences are enough.
   - [add] One sentence of lineage from the Discussion's second paragraph (L428–440) can move here.

2.1 Phase-Field Model of Dry Snow Metamorphism

1. **What a phase field is** (L191–198)

General description of the method.

- **[wording]** "representing the volume fraction of phase l at a given point ... and evolves through time": the sentence changes subject midway.
- **[trim]** Textbook material; one sentence.

2. **Unknowns** (L199–205)

   Lists the four fields.

   - **[correct]** "water vapor ϕ_a": ϕ_a is the air phase.
   - **[correct]** T is given in °C here and used as $T + T_0$ in Eqs. (3), (5); Table 1 and Eq. (7) need kelvin. Choose one.
   - **[wording]** "derived from the free energy potential and allows": "that allows".
   - **[trim]** The last two sentences ("defined pointwise", "benefit from the phase-field formulation") can go.

3. **Scope** (to write)

   Says which part of dry snow metamorphism the model contains.

   - **[add]** The model carries sintering by vapor transport: sublimation, diffusion through the pore, desublimation.
   - **[add]** Not in it: surface diffusion, grain-boundary processes, densification, overburden. Define desublimation.
   - **[add]** Hence the laboratory analogues are unloaded ice-sphere samples, not natural snow.

4. **Model equations and the phase equation** (L206–212)

   Eq. (1) and the meaning of its terms.

   - **[correct]** Eq. (1b): the latent-heat term should carry ρ_{ice}, as the solver does, not the mixture density, which integrates to about half across the interface.
   - **[correct]** "ε represents the interface width": it is the interface parameter; the visible band is $9.2\,\varepsilon$ (Table 3 says so).
   - **[correct]** Eq. (1c) as printed has no ξ_v, but Tables 3 and 4 list $\xi_v = 10^{-3}$. Show it or explain it in one sentence.
   - **[wording]** "Allen-Kahn" (Cahn); "localize this phenomena" (localizes this term); the sentence defining α_{sub} and ρ_i is run together.
   - **[add]** M_{sub} and α_{sub} are named and never given: $M_{\text{sub}} = \varepsilon/(3\tau_{\text{sub}})$, $\alpha_{\text{sub}} = \lambda_{\text{sub}}/\tau_{\text{sub}}$, or a pointer to Moure & Fu.

5. **Saturation vapor density** (L213–217)

   Eqs. (2) and (3).

   - **[correct]** Eq. (3): the last term reads $k_5 + \ln(T + T_0)$; the source form is probably $k_5 \ln(T + T_0)$. Check.
   - **[correct]** The text cites Flatau and Wexler; the note to Table 1 says ASHRAE.
   - **[wording]** "saturated vapor density of ice" (over ice); "ρ_a and P_a is" (are); Table 1 calls them ρ_{air}, P_{atm}.

6. **Energy equation** (L218–224)

   Eq. (1b) and the mixture rules.

   - [correct] "latent heat released and absorbed during sublimation and desublimation, respectively": reversed. Sublimation absorbs heat; desublimation releases it.
   - [wording] "phase dependent": hyphenate.

7. **Vapor equation** (L225–231)

   Eq. (1c) and the diffusivity.

   - [trim] T_0 is defined at L215 and again here.
   - [wording] "The rest of the thermophysical properties are shown in": "Material properties are listed in".

Table 1 [written] Material properties, cited at L230.

2.2 Interface Kinetics and the Sintering Time Scale

Draft heading: 2.2 Gibbs-Thomson Condition.

1. **The interface condition** (L232–242)

   Eqs. (6) and (7).

   - [wording] "is commonly interchanged with the condensation coefficient": "is commonly expressed through". "molecular mass a water molecule" (mass of a water molecule).
   - [add] $d_0 = \gamma m/(\rho_{\text{ice}} k_B T)$ is defined only in Table 1.

2. **The thin-interface limit** (to write)

   What "under the conditions outlined by Kaempfer & Plapp" means.

   - [add] The model recovers Eq. (6) with a kinetic coefficient $\beta_{\text{sub}} + \Delta\beta$, $\Delta\beta = a_1 a_2\, \varepsilon\, (D_T^{-1} + D_v^{-1})\, \rho_{\text{ice}}/\rho_{vs}^I$.
   - [add] Its size: 2% of β_{sub} in the aggregates, 0.2% in the saturated pair, 23% in the Molaro pairs.

   Cite: Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Moure & Fu 2024 `mourePhaseFieldModelWet2024`.

3. **The time scale** (to write here; now at L485–493 in the Discussion)

   Defines τ_{sub} and the sintering age θ.

   - [add] Move Eq. (τ_{sub}) from Section 5.1 to here, and define $\theta = t/\tau_{\text{sub}}$ here, not in Section 4.2.
   - [add] Two time scales of the same form $\ell^2\beta/d_0$: τ_{sub} ($\ell = \varepsilon$), the relaxation time of the diffuse interface; and $\tau_{\bar{R}} = \bar{R}^2\beta/d_0$, the time for a grain to change shape. $\tau_{\text{sub}} = \tau_{\bar{R}}/2500$ in every aggregate run.
   - [add] The 30-day runs end at $\theta = 43$ ($-40\,^\circ$C) to 1256 ($-5\,^\circ$C). Cite Table 2 here.

4. **The values used** (to write)

   The condensation coefficients and why they are constant.

- [add] $\alpha_c = 1.020 \times 10^{-3}$ (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair), constant in temperature.
- [add] Why constant: a temperature-dependent α_c would make a temperature sweep uninterpretable.

Cite: range of α_c: Braun et al. 2024 `braunRigorousApproachSpecific2024`; Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Libbrecht 2017 `libbrechtPhysicalDynamicsIce2017`.

Table 2 [written] Temperature-dependent quantities, placed. [add] Not cited in the text.

2.3 Effective Thermal Conductivity by Homogenization

Draft heading: 2.3 Homogenizing for Effective Thermal Conductivity.

1. **Two scales and the microscale problem** (L244–254)

   Scale separation, periodicity, Eq. (8).

   - [correct] "without less of generality": periodicity is an assumption, not a convenience (and the phrase is "loss").
   - [wording] "A smaller length scale l describing ..., and L describing ..." is not a sentence. "representative element volume" (elementary). "diffusive heat transfer" (conductive). "upscaling effective thermal conductivity" (computing the).

2. **Macroscale law and cell problem** (L255–263)

   Eqs. (9)–(11).

   - [correct] Eq. (11d) has $\mathbf{t}$ where it needs $\nabla\mathbf{t}$.
   - [wording] "are commonly referred to as co the solutions": the sentence is broken. "the corresponding macroscopic description is described as".
   - [trim] "Equations ... are then coupled with the boundary conditions ..." says the equations go with their own boundary conditions; cut.

3. **Single-domain form** (L264–276)

   Weak form and the indicator function, Eqs. (12)–(15).

   - [correct] The test function is scalar v in the text and bold $\mathbf{v}$ in Eq. (12). $K_\star$ in Eqs. (13)–(14) becomes K_* in Eq. (15).
   - [wording] "as diffuse interface model" (as a); "in 2.1" (Section 2.1); "We can the define" (then); "$\zeta = 1.0$" (1).
   - [trim] Eqs. (12)–(14) can be one sentence: summing the weak forms over the two subdomains, the interface terms cancel by flux continuity, and Eq. (15) follows.

4. **Diffuse form and the tensor law** (L277–291)

   Eqs. (16), (17) and why the conductivity is a tensor in the interface.

   - [correct] "interpolates between 0 and 1": between K_{air} and K_{ice}.
   - [wording] "along the our phase-field"; "$K_\parallel(\phi)$ and are $K_\perp(\phi)$ the"; "We then may write diffuse BVP" (the diffuse); "tonsorial" (tensorial); "interface width ε". ϕ and ϕ_i are used for the same field; $t_\star$ is not bold.

- [add] Credit and what is new: Nicoli, Plapp & Henry (2011) define the law and show the cancellation for a planar interface at contrasts of 2 and 10; here it is applied to the cell problem at a contrast of 115 and verified on laminate and disk-array cases (supplement).
- [add] Why it matters here: with a scalar law the bias reaches tens of percent and grows with ε SSA, so it would drift as the aggregate sinters.
- [add] The cell problem is solved on frozen snapshots of ϕ.

Cite: Nicoli, Plapp & Henry 2011 `nicoli_tensorial_2011`; Auriault 1991 `auriaultHeterogeneousMediumEquivalent1991`; Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

5. **What is reported** (to write)

   Defines the quantities the results plot.

   - [add] $\mathbf{K}_{\text{eff}}$ in the diffuse form, the one evaluated (Eq. 10 is the sharp form only).
   - [add] The scalar $k_{\text{eff}} = (k_{xx} + k_{yy})/2$, now defined only in the Figure 6 caption.
   - [add] SSA: how it is measured from ϕ, and its units. It is used throughout Section 4 and defined nowhere.

   Cite: for computing SSA from a field, possibly Flin et al. 2014 `flinComputationsSpecificSurface2014` (check that the method matches).

Figure 2 [written] placed. [add] Not cited in the text. [wording] Caption: "the mesh in which we solve our homogenization model over"; "red" lines are orange; panel (c) lacks its period; add that only the x problem is shown.

2.4 Numerical Implementation

1. **Discretization and time stepping** (to write; heading only at L292)

   Enough for someone to reproduce the runs.

   - [add] Quadratic C^1 splines; element size $\varepsilon/\sqrt{2}$.
   - [add] Adaptive step capped at $2\,\tau_{\text{sub}}$ in the aggregates. The grain-pair runs use larger caps (5 τ_{sub} for the saturated pair; 30 s, i.e. 22 τ_{sub}, for Molaro $-20\,^{\circ}\text{C}$); say so.
   - [add] Conductivity sampled every step for $\theta < 11$, then at every 0.1% drop in SSA.

2. **Checks** (to write)

   The results do not depend on domain size, time step or sampling.

   - [add] One sentence each, pointing to the supplement: domain size (Fig. S1); interface parameter (Fig. S2), including that the first day depends on ε and later rates do not; time step and sampling (Fig. S3).

Table 3 [written] Complete parameters, placed. [add] Not cited in the text. *Open: keep it here, or the short version here and this one in the supplement.*

2.5 Simulation Design: Grain Pairs and Aggregates

1. **Grain pairs** (to write; heading only at L293; material now at L317–329 and L342–347)

   What the three two-grain simulations are.

- [add] Axisymmetric about the line of centers; fixed temperature and humidity on the outer boundary; how the neck width w is measured.
- [add] The two Molaro geometries and the equal-grain pair ($R = 84.4$ μm, $-20\,^{\circ}$C, 100 h).
- [add] Every pair starts from two overlapping spheres with a neck 28 μm wide, not from grains that just touch. Say so here; it matters for the exponents (Section 3.2).

Table 4 [written] Two-grain parameters. [add] Add a column for the saturated pair. [correct] Simulated time at $-5\,^{\circ}$C is 68 min, not 2 h.

2. **Aggregates** (to write; material now at L359–368)

   What the 125 aggregate simulations are and how the packings were made.

   - [add] 2 mm square, periodic in both directions; circular grains, log-normal radii with mean 50 μm, placed by drop-and-roll deposition.
   - [add] Five porosities, five independent packings each, the same packings at all five temperatures: 125 runs of 30 days; uniform temperature, vapor initially at saturation.
   - [add] Domain size $L/\bar{R} = 40$, at the 1–2 mm representative volume of Calonne et al. (2011). Packings are accepted only if contacts across the periodic seam are as dense as in the interior.

   Cite: representative volume: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

3. **Why periodic, why two dimensions** (to write)

   Justifies the two choices a reviewer will question first.

   - [add] Periodic: the domain stands for the interior of a deposit, no wall physics has to be invented, and the system is closed, so curvature alone drives transport.
   - [add] Two dimensions: cost (now at L306–307, in the wrong place); what it costs is discussed in Section 5.6.

3 Neck Growth Between Two Ice Grains

1. **Section opening** (L295–310)

   Announces the findings.

   - [correct] It was written as the opening of a single Results section and still announces the aggregate findings of Section 4.
   - [correct] "independent of the mean temperature as well as the initial porosity"; "our model's ability to capture sintering dynamics for varying temperatures": neither is what the results show.
   - [correct] "theorized trends from Demmenie et al.": they measured the exponent as well. "dominated by vapor transport": the $t^{1/3}$ law is for sublimation and desublimation limited by attachment.
   - [wording] "simulation results, which captures" (capture); "related the evolution" (to the); "microstructures SSA" (microstructure's).
   - [trim] Two sentences: before using the model on aggregates, we ask how much of measured neck growth it reproduces, with one experiment away from saturation and one at saturation.

3.1 Comparison With Unsaturated Experiments

1. **Why these experiments** (L311–316)

   Why Molaro et al. is the comparison.

   - [correct] "relatively well-controlled": the humidity was not reported and the grains sublimated; Section 5.1 argues the conditions were not controlled at the level of the driving force. Say "among the few that isolate a grain pair at these sizes and temperatures".
   - [wording] "are good candidate" (a good candidate); "validating" (testing); "that exist that isolates" (isolate).

2. **What is fitted and what is not** (L317–329)

   The set-up of the $-20\,^{\circ}C$ run.

   - [correct] "The only free-parameter in our model is ... β_{sub}": the humidity is fitted too.
   - [correct] "over a 78 minute period": Molaro's text describes their Figure 11 as 57 min apart; our digitized series runs to 78 min. Check which figure the series comes from and cite it.
   - [correct] "within a sealed cryostage": their text says the sample area was not sealed off from the chamber, and the grains shrank.
   - [wording] "fig. 11" (their Figure 11); "We assumed a value ... and was at the upper end" (state $\alpha_c = 0.1$); "(cite other works as well)"; "table table 4".
   - [add] The inferred humidity, 0.99715 at $-20\,^{\circ}\mathrm{C}$ and 0.99674 at $-5\,^{\circ}\mathrm{C}$, from the shrinkage of the large grain.
   - [trim] Geometry and boundary conditions move to Section 2.5.

3. **Result** (L330–340)

   Figure 3: model against measurement.

   - [wording] "Panel (a) we see" (shows); "the a digital replicate"; "corresponds final temporal data point"; "over approximates and under approximates" (overpredicts, underpredicts); "will discussed" (be); "section section 5.1"; "that they cryostage chamber".
   - [add] The numbers: 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$ over 78 min; 170% at $-5\,^{\circ}\mathrm{C}$ over 48 min (52.1 against 44.0 μm).
   - [add] Panels (c), (d): the shrinkage is what the humidity was fitted to, so they are a calibration, not a test.
   - [add] Molaro's conventions are used (neck width; time from the first measurement). Their own model matches through its vapor term, which assumes $\alpha_c = 1$ and no gas diffusion.

   Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

Figure 3 [written] placed; caption now covers (c), (d). [wording] "filed" (field); "the the". [correct] (c), (d) show model and data, not only the "experimentally measured grain size".

3.2 Growth at Saturation

1. **Set-up and result** (L342–356)

   The saturated equal-grain run and its fitted exponent.

- **[correct]** "a vapor-limited regime … governed by vapor transport": the reverse. At $\alpha_c = 10^{-3}$ growth is limited by attachment (gas diffusion adds 0.4–2% to the resistance), which is the regime the $t^{1/3}$ law describes.
- **[correct]** "two ice grains briefly brought into contact": the run starts from a neck 28 μm wide.
- **[correct]** "The red dashed curve is the theoretical fit": it is the best fit with the exponent fixed at 1/3.
- **[wording]** "in attempts to compare" (to compare); "allowed for the to sinter"; "an condensation"; "$1.002e-03$" (1.002×10^{-3}); "Similarly to".
- **[add]** Why this run: at saturation the grains do not shrink (radius +0.06%), so neck growth is isolated. Neck from 28 to 63 μm.
- **[add]** Demmenie et al. measure 0.26–0.33 at saturation and about 1/7 when undersaturated. The fitted exponent rises from 0.21 to 0.24 as the window starts later. The unsaturated $-20\,^{\circ}$C run gives 0.135 (Molaro's data 0.23 ± 0.03). No exponent at $-5\,^{\circ}$C.

Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`.

2. **Why the exponent is low** (to write; provisional until the tangent-start run is in)

   The sharp-interface law gives 0.29 for this run; the 0.21 comes from the neck it was started from.

   - **[add]** The Gibbs–Thomson prediction for this geometry and these parameters: exponent 0.29, final neck width within 4%.
   - **[add]** The starting neck is two overlapping spheres: a sharp notch, with the centers 2.3 μm closer than touching grains.
   - **[add]** The run grows faster than the prediction at first (2.5 times at 2 h) and settles onto it (1.0 by 71 h).
   - **[add]** The fit must assume how long the neck had been growing: a free fit takes 4.8 h and gives 0.21; the prediction says 16 h, and with that the same data give 0.28.
   - **[add]** The run from grains just touching: *fill in from the pending run.* (The August run of that kind, at twice the interface width, gave 0.283 with no early excess.)
   - **[add]** The limit of the claim: within 100 h the rate reaches the prediction but the fitted exponent does not (0.236 against 0.29).

   Open: if the pending run still shows the early excess at equal neck widths, the cause is the interface width instead. A claim of convergence as $\varepsilon \to 0$ is not available without a run at smaller ε.

Figure 4 **[written]** placed. **[wording]** Caption: "shown undergoing sintering shown"; "filed"; "the the". **[add]** Temperature, grain size, the fit window (1–100 h).

4 Thermal Conductivity of Sintering Aggregates

4.1 Microstructural Evolution

1. **Set-up and the gallery** (L359–377)

   States the simulation matrix, then describes Figure 5.

- [correct] "mesh of size 2829×2928": 2829 both ways.
- [correct] "We initially consider a single 2D microstructure": the previous sentence states 125 runs, and Figure 5 shows five.
- [correct] "periodic boundary conditions for our equation (1b) and (1c)": all three fields are periodic.
- [correct] "By the 10 day snapshot for each porosity ... a well connected icy porous matrix": the ice already spans the cell at $t = 0$ in 24 of 25 packings; what changes is that point contacts become necks. At the two highest porosities the network stays anisotropic.
- [wording] "Next, we aim to better understand ... to better understand" (twice); "five different configurations for five separate porosities" (five packings at each of five porosities); "drop and roll" (hyphens); "gravitational affects" (effects); "a a".
- [trim] The set-up (first half) moves to Section 2.5.
- [add] What the colors show: vapor is undersaturated over convex ice, which sublimates, and supersaturated in necks and concavities, where it desublimates.

Figure 5 [written] placed, with caption.

4.2 Temperature Rescales Time

2. **Computing the conductivity** (L378–387)

   Applies the homogenization and describes the tensor.

   - [correct] "We obtain ... $\mathbf{K}_{\text{eff}}$ by integrating (17a) and applying the zero mean condition": it is the volume average of $\mathbf{K}(\nabla \mathbf{t} + \mathbf{I})$ over the cell (Eq. 10 in diffuse form).
   - [correct] "off diagonal components are at least one order of magnitude lower": k_{xy} reaches about 12% of k_{xx} on one packing; it is zero in the mean.
   - [wording] "changes ... is expected" (are); "figure figure 6"; "correspond the the"; "lower that" (than); "over various seeds" (packings).
   - [trim] The two sentences on solving Eq. (17) on the same mesh repeat Section 2.3.
   - [add] Anisotropy: $k_{xx}/k_{yy} = 1.02$, 1.03, 0.93, 0.82, 0.66 from the lowest porosity to the highest, with y the direction of deposition. The scalar k_{eff} is used from here on.

3. **Figure 6, one packing at five temperatures** (L388–397)

   Conductivity and SSA against time and against sintering age.

   - [correct] "After the first day": that is 11 τ_{sub} at $-20\,^{\circ}\text{C}$ only; the relaxation period is a fixed number of τ_{sub}, so it is 0.26 d at $-5\,^{\circ}\text{C}$ and 7.7 d at $-40\,^{\circ}\text{C}$.
   - [correct] "the trend is linear": linear in $\log\theta$, about 0.13 per decade.
   - [wording] "local sintering occurs at high rates ... and the phase-field interface relaxes" joins two different things; "non-dimensional" here, "nondimensional" in the caption.
   - [add] Panel (a) is not mentioned.
   - [add] The result as a number: against θ the rates follow τ_{sub} within 4%; against SSA the five temperatures agree within 0.27% for every one of the 25 packings (median 0.07%).
   - [add] A size that does not depend on ε: between $\theta = 30$ and 331, k_{eff} rises by a factor 1.171 ± 0.015 and SSA falls to 0.819 ± 0.007, for $\varphi \leq 0.375$.

- [add] Why curves are normalized at $\theta_r = 30$ and not at $t = 0$: the first day depends on the interface parameter (Fig. S2).

Cite: for SSA against log time in isothermal snow: Legagneux & Dominé 2005 `legagneuxMeanFieldModel2005`; Taillandier et al. 2007 `taillandierRateDecreaseSpecific2007` (functional form only; verify).

Figure 6 [written] placed. [correct] Caption: $|\sigma| \leq 0.34$ (it says 0.31). [wording] "preformed" (performed); "The vertical axis is ..., is normalized"; "2.0mm". [add] Define θ_r.

4.3 Conductivity as a Function of Specific Surface Area

4. **Figure 7, porosity** (L398–416)

   One paragraph covering all four panels.

   - [correct] "15 unique packings–3 packings for each porosity": five packings at each of three porosities.
   - [correct] "We show the time evolution of k_{eff} for each of the simulations": panel (a) is $-20\,^{\circ}\text{C}$ only, the mean of five packings and their range.
   - [correct] "the ice matrix remains disconnected in the horizontal direction": not so. The ice spans the cell in both directions in every packing from 10 d on (percolation test). What happens is that the packings become anisotropic and the two directions follow different laws: along the deposition direction the exponent stays at -0.79 to -0.86 up to $\varphi = 0.425$; across it, -0.70, -0.60, -0.48 from $\varphi = 0.375$ on.
   - [correct] "The linear trend" in panel (c): the axis is logarithmic; the fit is exponential in porosity.
   - [wording] "may effect" (affect); "scientifically their connectivity as." is unfinished; "the strength of the reference value" (the reference value); "each of the the"; "A single point ... show" (shows); "panels(c)".
   - [trim] One paragraph carries four panels; split into three: (a); (b); (c) and (d).
   - [add] Why four temperatures: the $-40\,^{\circ}\text{C}$ runs end at $\theta = 43$, too soon after the reference age.
   - [add] The numbers. Exponent -0.80; by porosity -0.79, -0.82, -0.78, then -0.73, -0.63. Level $k_{\text{eff},r}/k_{\text{ice}} = 1.37\exp(-5.01\,\varphi)$ for $\varphi \leq 0.375$, 6.6% scatter between packings. At 30 d the conductivity differs by a factor of about 3.3 across porosity.
   - [add] What it amounts to: SSA gives the relative change in conductivity; porosity gives the level; neither alone gives both.

   Open: whether to plot or tabulate the directional exponents; whether to add that a porosity-dependent representative volume may contribute (not tested: the domain-size study was run at $\varphi = 0.325$ only).

Figure 7 [written] placed. [correct] Caption (a): the mean of five packings and their range, not "a single seed"; (c), (d): one point per packing, not per simulation. [wording] "Plots the the". [add] The arrow and the gray band.

5. **What the scatter is** (optional; to write)

   Differences between packings in the exponent are properties of the packings, not noise.

- **[add]** The scatter in the exponent does not fall with domain size while the scatter in the level does (11% to 4.5%).

Open: include here, in Section 5.3, or leave out.

5 Discussion

Paragraphs 1–3 of Section 5.1 and the start of Section 5.2 are drafted.

(a) **Section opening** (L418–427)

Recaps the paper.

- **[correct]** "our model is able to capture the trends in sintering dynamics at multiple temperatures": at $-20\,^{\circ}\mathrm{C}$ it gives 41% and at $-5\,^{\circ}\mathrm{C}$ 170%.
- **[trim]** The third statement of the plan (after the Introduction and the opening of Section 3). Not needed.

(b) **Lineage of the model** (L428–440)

Kaempfer & Plapp, Moure & Fu, and what this paper adds.

- **[wording]** "They also, qualitatively validate"; "has since become popular method".
- **[trim]** Belongs in the Methods opening, in one or two sentences.

5.1 What Vapor Transport Alone Reproduces

1. **The unsaturated pair** (L442–468)

Why the model gives 41% of Molaro's neck growth and what that does and does not mean.

- **[correct]** "underpredicts … overpredicts … by 41% and 170%": the model gives 41% *of* the measured growth at $-20\,^{\circ}\mathrm{C}$ (it underpredicts by 59%) and 170% of it at $-5\,^{\circ}\mathrm{C}$.
- **[correct]** "It does not account for surface diffusion, which is expected [to] contribute significantly" (twice): by Molaro's own equations surface diffusion is below 1% of vapor transport at these neck sizes (Appendix A). Remove.
- **[correct]** "This result was expected": it was not.
- **[correct]** "The chamber being unsaturated … [is] likely the reason why our model predictions overestimate": the undersaturation is already in the model, through the inferred humidity.
- **[wording]** "preformed" (performed); "the the"; "our model own accounts" (only accounts); "expected contribute"; the three sentences from "Furthermore, our model also underestimates" appear twice.
- **[trim]** The first three sentences repeat Section 3.1.
- **[add]** "Even with the vapor diffusivity raised 30 or 100 times": say these were tests not shown, and that the neck was insensitive to the chamber humidity at the nominal diffusivity.
- **[add]** How small the driving force is: 4.8×10^{-4} falling to 9×10^{-5} in relative vapor density, the equivalent of 4.9 to 1.0 mK; the chamber's undersaturation is 2.85×10^{-3}, 6 to 31 times larger.

- [add] What that makes the result sensitive to, without choosing: (i) how much of the ambient undersaturation reaches the neck (about 6% in our run; with none, the one-equation neck model gives 66% of the growth and an exponent of 0.23, Molaro's value); (ii) temperature differences of millikelvin between neck and grain surface; (iii) α_c above 0.1 for droplets quenched in liquid nitrogen.
- [add] Latent heat is not a candidate: in the run the neck is 0.002 mK warmer than the grain surface.
- [add] Molaro's own model matches through a vapor term with $\alpha_c = 1$ and no gas diffusion, and they present its agreement as order-of-magnitude.
- [add] Why the saturated experiment is the better test: it removes the undersaturation.

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`; Swinkels & Ashby 1981 `swinkelsSecondReportSintering1981`.

2. **The saturated pair, and exponents** (L469–484; provisional until the tangent-start run is in)

   The model follows the saturation trend in the exponent; where it falls short, the cause is the starting neck.

   - [correct] "As two spherical ice grains sinter in an environment with a saturated water vapor field, the growth is expected to be kinetics limited": saturation does not make growth attachment-limited; a small α_c does. The $t^{1/3}$ law assumes it.
   - [correct] "a much better match for the theorized scaling … $t^{1/7}$": say who reports 1/7 for undersaturated conditions (Demmenie et al.) and that it is an observation there.
   - [correct] "they are lower than measured values": give the pairs, 0.21 against 0.26–0.33 at saturation and 0.135 against 0.23 ± 0.03 at $-20\,^{\circ}\mathrm{C}$.
   - [wording] The first sentence is unfinished ("growing under as ."); "where observe"; "we had considered"; "$1.002e-03$"; "preform" (perform); two sentences in a row begin "We additionally" and "Additionally".
   - [add] "We additionally compare our measured interface velocity …" states no result: 2.5 times the Gibbs–Thomson value at 2 h, 1.0 by 71 h; the prediction's exponent is 0.29.
   - [add] The reason for the shortfall, from Section 3.2, in two sentences. The Molaro runs start from the same pre-made neck and their comparison begins about 100 τ_{sub} after the start.
   - [add] Why not surface diffusion: mechanisms in parallel add their rates, and surface diffusion alone gives $t^{1/7}$, so adding it could not raise an exponent toward 1/3.

   Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`; Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

3. α_c **as an effective parameter** (L485–493)

   What the uncertain coefficient does and does not change.

   - [correct] "do not change how the microstructure evolves, just the rate": true only while growth is limited by attachment. α_c sets the time scale ($\tau_{\mathrm{sub}} \propto 1/\alpha_c$) *and* the group $\Lambda = L^*/\bar{R} \propto 1/\alpha_c$ that decides how much gas diffusion matters. In the aggregates $\Lambda \approx 2.8$. A smaller α_c only rescales time; a larger one brings diffusion in and changes the path as well. (An earlier version of this outline said a tenfold larger α_c would leave the curves unchanged against θ; that was wrong.)

- [correct] "$\Delta\beta/\beta_{\text{sub}} = 0.02$" for both the aggregates and the saturated pair: it is 0.02, 0.002, and 0.23 for the Molaro pairs.
- [correct] The sentence on Bouvet et al. rescaling time through α_c cites the 2023 paper; check that it is the one (the fit of α_c to tomography is Bouvet et al., 2022).
- [wording] "The kinetic attachment parameter, and thus condensation coefficient, do not"; "μm-CT" (μCT).
- [trim] Eq. (τ_{sub}) and the sentence after it move to Section 2.2.
- [add] What α_c does to the two grain-pair comparisons: Molaro's rate wants a large value (41% at 0.1; 51–75% at 0.8 in the one-equation neck model), while the sharp-interface exponent at saturation falls with it (0.29 at 10^{-3}, 0.25 at 10^{-2}, 0.18 at 0.1). One value cannot serve both.
- [add] [from the notes] Bouvet et al. fit $\alpha_c = (9.8 \pm 0.7) \times 10^{-4}$ to tomography at $-2\,^{\circ}\text{C}$; Braun et al. find 10^{-3}–10^{-1}, changing as surfaces facet, and call it an effective, surface-averaged value.

Cite: Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Braun et al. 2024 `braunRigorousApproachSpecific2024`; Granger et al. 2021 `grangerOrientationSelectiveGrain2021` for faceting under a gradient; Libbrecht & Rickerby 2013 `libbrechtMeasurementsSurfaceAttachment2013` as background on why α_c is uncertain.

5.2 Temperature as a Rescaling of Time

4. **Why the curves superpose** (L495–503; breaks off at "Since we have already")

   A similarity argument: in dimensionless form, almost all of the temperature dependence is in one time scale.

   - [correct] "β_{sub}, which is a free variable": it is set by α_c, which is chosen, and it depends on temperature through $\rho_{vs}^{I}(T)$. That dependence is the point of the paragraph.
   - [wording] "is evident … and is evident"; "a two part series" (two steps in series); "drawn to areas of lower vapor saturation".
   - [add] Step 1. Scale lengths by $\bar{R}$, supersaturation by $d_0/\bar{R}$, time by $\tau_{\bar{R}} = \bar{R}^2\beta/d_0$ ($= 2500\,\tau_{\text{sub}}$). The interface condition becomes parameter-free. The saturation vapor density, with its 31-fold change from -40 to $-5\,^{\circ}\text{C}$, is entirely inside the time scale.
   - [add] Step 2. One temperature-dependent group is left, $\Lambda = L^*/\bar{R}$ with $L^* = (D_v/\alpha_c)\sqrt{2\pi m/k_B T}$. The saturation vapor density cancels in it; what remains is the diffusivity. It changes by 20% where the time scale changes by a factor of 29.
   - [add] Step 3. So the saturation vapor density sets the clock, and the diffusivity is a weak remaining correction: the likely source of the 4% by which rates depart from τ_{sub} (consistent in size; not tested as the cause).
   - [add] Say plainly that with one sintering mechanism this is close to a consequence of the construction; the numerical result is how well it holds. Do not say gas diffusion is negligible ($\Lambda \approx 2.8$); what is small is its change with temperature.
   - [add] The two-page derivation (`temperature_scaling/time_temperature_scaling.pdf`) can be an appendix or supporting information.

5. **What would break it, and the activation energy.** (new) [to write]

Message: *The collapse is therefore a signature of a single route. Its temperature dependence differs from the one measured for strength gain, which points to a second route in real samples and gives experimenters something to measure.*

The clock's temperature dependence is an activation energy of about 50 kJ mol^{-1}, that of the saturation vapor density. **[from the notes]** Choukroun et al. measure 24.3 ± 3.3 kJ mol^{-1} for strength gain and attribute it to surface self-diffusion (L613–620), at 193–243 K; the two temperature ranges overlap at 233–243 K. The difference is consistent with a second route in their samples, and an activation energy is something a laboratory can measure to tell the routes apart. A temperature-dependent α_c, a second route, or a temperature gradient would each break the superposition, which makes it a test.

Cite: Choukroun et al. 2020 `choukroun_strength_2020`.

5.3 Specific Surface Area as the State Variable

6. **Density is not enough.** (new) **[to write]**

   Message: *Conductivity changes while porosity stays fixed, so a density-based formula cannot describe a sintering deposit.*

   In these runs porosity is fixed and conductivity still rises, by 17% between $\theta = 30$ and 331. **[from the notes]** Calonne et al. (2011): tensor from 30 tomographic images, a regression on density, scatter attributed to anisotropy (L507–517). Calonne et al. (2014): under a temperature gradient conductivity changes at constant density, with SSA and anisotropy (L517–529). Half the length of the notes.

   Cite: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`; Calonne et al. 2014 `calonneStudyTemperatureGradient2014`; Schneebeli & Sokratov 2004 `schneebeliTomographyTemperatureGradient2004` (the notes cite it for the same behavior under a gradient).

7. **The power law and where it stops.** (new) **[to write]**

   Message: *State the law, its range, and that outside the range we make no claim.*

   $p = -0.80$ for $\varphi \leq 0.375$; the level from porosity; above 0.375 the exponent weakens and the packings are anisotropic. No claim that packings of different porosity converge.

   Cite: Calonne et al. 2011, 2019 `calonneNumericalExperimentalInvestigations2011`, `calonneThermalConductivitySnow2019` for conductivity against density, as the contrast.

8. **Why packings differ.** (new) **[to write]**

   Message: *Packings of the same porosity conduct differently because their ice networks are arranged differently; at high porosity the networks are anisotropic and the simple law fails in one direction.*

   Packings of one porosity differ in how well the ice is connected, and the exponent follows that (correlation -0.77 with conduction efficiency), not coordination number or anisotropy. In these packings the ice always spans the cell and the pore never does (percolation test, 2026-10-12): the pore space is a set of closed pockets, which is what "sealed pores" means here, and in two dimensions only one phase can span. What differs between packings is the quality of the ice network, not whether it exists: at high porosity the packings are anisotropic, grains dropped under gravity being supported from below with fewer contacts sideways, and the power law survives along the deposition direction while failing across it. Absolute values are therefore not compared with three-dimensional measurements, where both phases are connected. *Open:*

literature thresholds for discs are not yet checked against a source; whether to show the efficiency; the Euler-characteristic series (tested on three runs) is available if a direct measure is wanted.

5.4 What a Laboratory Could Measure

9. **Three tests.** (new) **[to write]**

   Message: *Turn the results into measurements someone could make on sintered ice-sphere samples.*

   On unloaded ice-sphere samples held isothermally: (i) curves at different temperatures superpose under a time shift, and the shift gives an activation energy; (ii) conductivity against SSA is one curve; (iii) SSA falls linearly in log time. **[from the notes]** SSA is measured by near-infrared reflectance, gas adsorption and tomography (L562–568; Gallet, Matzl, Proksch).

   Cite: Gallet et al. 2009 `galletMeasurementSpecificSurface2009`; Matzl & Schneebeli 2006 `matzlMeasuringSpecificSurface2006`; Proksch et al. 2015 `prokschDensitySpecificSurface2015`; for isothermal SSA decay in the laboratory, Legagneux & Dominé 2005, Taillandier et al. 2007, Kaempfer & Schneebeli 2007 `kaempferObservationIsothermalMetamorphism2007`, Löwe et al. 2011 `loweInterfacialStructuralRelaxations2011` (forms only).

5.5 Implications for Plume Deposits on Enceladus

10. **What carries over.** (new) **[to write]**

    Message: *For Enceladus the qualitative conclusion holds: a sintering deposit conducts better as its surface area falls, even with no compaction.*

    Qualitative: as a deposit sinters, SSA falls and conductivity rises together, without densification. **[from the notes]** Tiger Stripe temperatures up to about 180 K (Spencer); grain size increasing toward the fractures in VIMS data (Jaumann; L580–602); Molaro et al. on the dependence of sintering on temperature and grain size.

    Cite: Spencer et al. 2009 `spencerEnceladusActiveCryovolcanic2009`; Spencer & Nimmo 2013 `spencerEnceladusActiveIce2013`; Jaumann et al. 2008 `jaumannDistributionIcyParticles2008`; Molaro et al. 2019; Choukroun et al. 2020.

11. **Why it matters, and the gap.** (new) **[to write]**

    Message: *Conductivity of the surface layer affects thermal and radar interpretation, but our conditions are far from the moon's, so we give no timescales.*

    [from the notes] A low-conductivity surface layer changes subsurface temperatures and radar attenuation (Byrne; L603–612); Choukroun et al. on strength gain and fallout rates of 0.01–1 $\mathrm{mm\,yr^{-1}}$ (L613–629). Then the gap, stated plainly: the runs stop at $-40\,^{\circ}\mathrm{C}$ and use 50 μm grains in air; the moon is far colder, the grains are micron-sized, and there is no air. No timescales are given.

    Cite: Byrne et al. 2024 `byrneRadarAttenuationSubsurface2024` (a preprint DOI in the `.bib`; check whether it is published); Choukroun et al. 2020; possibly Ferrari et al. 2021 `ferrariThermalEmissionSaturns2021` on thermal emission of Saturn's icy moons (in your `.bib`; check relevance).

5.6 Limitations

12. **Model.** (new) **[to write]**

    Message: *What physics is missing.*

Vapor route only; constant α_c; air-filled, sealed pores at 1 atm; conduction only and constant k_{ice}, k_{air}. **[from the notes]** Fourteau et al. on the contribution of vapor transport to apparent conductivity under a gradient (L569–575; check the figure quoted); Calonne et al. (2019) on temperature-dependent conductivities (L575–579).

Cite: Fourteau et al. 2021 — your `.bib` has two, `fourteauImpactWaterVapor2021` and `fourteauMacroscopicWaterVapor2021`; the point about apparent conductivity is the first. Calonne et al. 2019 `calonneThermalConductivitySnow2019`.

13. **Numerics and geometry.** (new) **[to write]**

    Message: *What the simulations cannot resolve or represent.*

    Two dimensions, so relative changes are claimed and absolute conductivities are not; one grain size distribution; the first day depends on the interface parameter, and the grain-pair runs start from a neck put in by hand, which grows faster than the Gibbs–Thomson law for the first two thousand τ_{sub} or so (Section 3.2), so exponents fitted there are biased low; the Molaro runs use time steps of 22 and 49 τ_{sub}, outside the range the time-step study covered; pore throats narrower than the diffuse band (9.2 μm) are not resolved, which raises k_{eff} most at low porosity; the power law is established for $\varphi \leq 0.375$; rates scale with $1/\alpha_c$, whose value is uncertain over two decades.

    Cite: none needed beyond those above.

In the notes and not to be carried over: $\beta_{\text{sub}}(T)$ from Libbrecht's $\sigma_0(T)$, a single tuning and a "genuine prediction" at $-5\,^{\circ}\text{C}$ (L454–481); isotropy, and a collapse across porosity (L530–561); sintering "on the order of decades" (L596) and conductivity starting "near 0.0" (L611).

6 Conclusions

1. **What was done and found.** (new) **[to write]**

   Message: *Three results, each with the range over which it was shown.*

   One sentence of method, then the three results with their ranges: temperature rescales time (-40 to $-5\,^{\circ}\text{C}$, within 0.3%); conductivity follows $\text{SSA}^{-0.80}$ with the level set by porosity ($\varphi \leq 0.375$); vapor transport gives 41% of measured neck growth at $-20\,^{\circ}\text{C}$.

2. **What comes next.** (new) **[to write]**

   Message: *The experiment and the model extensions that would test or extend them.*

   The laboratory test of Section 5.4; a second transport route in the model; three dimensions; colder temperatures and smaller grains.

Open Research; Acknowledgments

[to write] Code and data availability; run inventory (`RUN_TABLE.md`). Acknowledgments.

Supporting Information

- **Figures.** S1 domain-size convergence; S2 interface parameter; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.

- **Tables, text.** Complete parameters, if moved; packing construction; the weak-form derivation, if trimmed; run table.

Appendix: the two calculations behind Sections 3 and 5.1

A. Surface diffusion against vapor transport, by Molaro's own equations

Question. Does surface diffusion supply the 59% of neck growth our model lacks at −20 °C?

Equations (Molaro et al., 2019, Table 3 and Eqs. 1–4, 6, 17–20; a grain radius, x neck radius, S their smoothing factor):

$$\dot{V}_1 = S\,\frac{3\pi x\,\delta_s D_s\,\gamma_s\Omega\,(K_3 - K_2)}{d_2\,kT} \qquad \text{surface diffusion}$$

$$\dot{V}_3 = S\,2\pi x\rho\theta\,P_v\,\frac{\gamma_s\Omega}{kT}\left[\frac{\Omega}{2\pi\Delta_o kT}\right]^{1/2}(K_3 - K_m) \qquad \text{vapor transport}$$

$$\dot{x} = \frac{1}{2\pi x\theta\rho}\sum_i \dot{V}_i$$

$$K_3 = \frac{2}{a},\quad K_m = -\frac{1}{\rho} + \frac{1}{x},\quad \rho = \frac{x^2}{2(a-x)},\quad \theta = \tan^{-1}\frac{a}{x+\rho},\quad K_1K_2 = K_m^2,$$

$$\frac{K_2 - K_1}{K_1} = \frac{3A\rho}{x} + \frac{3}{2} - \frac{3}{2}\left[3\Big(\frac{A\rho}{x}\Big)^2 + \frac{4A\rho}{x} + 1\right]^{1/2},\quad d_2 = \frac{\rho}{1 + \left(\frac{4(K_2-K_1)}{3K_2}\right)^{1/2}},\quad A = \frac{\delta_b D_b}{\delta_s D_s}.$$

The vapor term is Hertz–Knudsen with $\alpha = 1$ and no gas diffusion (their Appendix A).

Parameters (their Table 5): $\Omega = 1.181\times10^{-29}$ m^3; $\Delta_o = 917$ kg m^{-3}; $\gamma_s = 0.06$ J m^{-2}; $\delta_s = \delta_b = 9.04\times10^{-10}$ m; $D_{0s} = 1.4\times10^{-8}$ m^2 s^{-1} (literature range down to 1.35×10^{-13}), $Q_s = 23.2$ kJ mol^{-1}; $D_{0b} = 8.4\times10^{-4}$ m^2 s^{-1}, $Q_b = 4.9$ kJ mol^{-1}; P_v from Murphy & Koop (2005), 103 Pa at −20 °C.

At their experiment ($a = 87$ μm, −20 °C; measured mean $\dot{x} = 3.4\times10^{-9}$ m s^{-1}):

x [μm]	ρ [μm]	$\dot{V}_1/\dot{V}_3$	$\dot{x}$ from $\dot{V}_1$ [m/s]	$\dot{x}$ from $\dot{V}_3$ [m/s]	$\dot{V}_1/\dot{V}_3$, $A = 0$
16.4	1.9	6.7×10^{-3}	5×10^{-11}	7.8×10^{-9}	8×10^{-7}
20.0	3.0	3.1×10^{-3}	2×10^{-11}	4.9×10^{-9}	3×10^{-7}
25.0	5.0	1.3×10^{-3}	4×10^{-12}	2.9×10^{-9}	1×10^{-7}
32.4	9.6	4.1×10^{-4}	6×10^{-13}	1.5×10^{-9}	4×10^{-8}

- Surface diffusion is 0.7% falling to 0.04% of their vapor term. It exceeds it only for x below about 4 μm.
- Their vapor term alone gives the measured rate.
- The last column switches off the coupling to grain-boundary diffusion ($K_2 = K_m$). With their parameters $A = 3.5\times10^8$, and the coupling raises the driving curvature K_2 by a factor of about 4000.

Check and its limit. Their Figure 10 (25 μm, $T/T_m = 0.8$) was recomputed. The vapor current agrees (2×10^{-21} rising to 3×10^{-20} m^3 s^{-1}; theirs about 10^{-21} to 2×10^{-20}). The surface current starts about 100 times above theirs, because they solve the four coupled equations iteratively where

this uses the approximate Eq. (20). The table therefore *overstates* surface diffusion. Script and output: `studies/molaro_2019/sa81_mechanisms/`.

What it implies. The gap is between our vapor route ($\alpha_c = 0.1$, diffusion through air) and whatever supplies vapor in the experiment. A one-equation neck model fitted to the same data wants $\alpha_c \approx 0.3$–1 (`studies/molaro_2019/envelope_ode/`).

B. The saturated pair against the Gibbs–Thomson law

Question. Is the exponent of 0.21 physics or resolution?

Prediction. For a neck of radius x between grains of radius R, with fillet radius $\rho = x^2/(2(R-x))$, fed from the grain surface through attachment in series with gas diffusion over a length $c\rho$:

$$v_n = \frac{d_0 \left(\dfrac{1}{\rho} - \dfrac{1}{x} + \dfrac{2}{R} \right)}{\beta_{\mathrm{sub}} + c\,\rho\,k_D}, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\pi}{2}\, v_n, \qquad k_D = \frac{\rho_{\mathrm{ice}}}{D_v\, \rho^I_{vs}}.$$

Attachment alone gives $x^3 \propto t$; diffusion alone $x^5 \propto t$. $R = 84.4$ μm, $-20\,^\circ$C, $\alpha_c = 10^{-3}$; $c = 0.275$ from the Molaro run, not refitted. Here gas diffusion is 0.4–2% of the resistance. Integrated from the run's first sample after relaxation, it gives a final width of 60.6 μm (run: 63.2) and an exponent of 0.29 (run: 0.21).

Run against prediction ($\varepsilon = 0.12$ μm; one interface band $= 9.19\,\varepsilon = 1.1$ μm; exponents from fits of $C(t + t_0)^a$ starting at the time shown):

Start	w [μm]	w/ε	fillet [bands]	velocity, run / prediction	a, run	a, prediction
2 h	35.1	293	2.1	2.54	0.215	0.292
6 h	39.2	327	2.7	1.72	0.221	0.292
16 h	44.9	374	3.7	1.36	0.227	0.290
31 h	50.1	418	4.8	1.18	0.231	0.288
51 h	55.0	459	6.0	1.08	0.234	0.287
71 h	58.8	490	7.1	1.02	0.236	0.286

The excess fits $1 + 11.4\,(\mathrm{band/fillet})^{2.7}$ (rms 0.035).

Sensitivity to the assumed contact time (same data, t_0 fixed):

t_0	4.8 h (free fit)	8 h	12 h	16 h	24 h
exponent	0.212	0.234	0.258	0.281	0.325
rms misfit	0.28%	0.78%	1.33%	1.73%	2.31%

The prediction takes 16 h to grow a neck to the run's first fitted width.

What it shows and what it does not. The rate converges on the sharp-interface law, and the early excess accounts for most of the difference in exponent. Two causes fit these numbers equally: the interface being too wide for a small fillet, or the pre-made neck relaxing from its imposed shape. The August run from tangent contact, at twice this interface width, showed no excess and gave 0.283, which points to the pre-made neck. The pending run from contact at this interface width tests it directly. Neither shows the predicted exponent being reached within 100 h from a pre-made neck (0.236 at the end against 0.29). Scripts and output: `studies/molaro_2019/demmenie/` (`predict_neck_growth.py`, `interface_width_estimate.py`).

C. What the appendices of the manuscript should hold

The draft now has two appendix headings with these titles. Suggested content, each under a page:

- *Appendix A, Surface diffusion against vapor transport:* the two rate equations and the curvature relations, the parameter list, the four-row table, and the two sentences on what was and was not reproduced of Molaro's Figure 10.
- *Appendix B, The saturated pair against the Gibbs–Thomson law:* the neck law, the run-against-prediction table, the contact-time table, and the result of the tangent-start run.
- The time–temperature scaling (two pages, `temperature_scaling/time_temperature_scaling.pdf`) is a candidate third appendix, or supporting information, for Section 5.2.

Citations: what the .bib lacks

Suggested above and present in the `.bib`: all of them. Mentioned in this plan and *not* in the `.bib`:

- Auriault, Boutin & Geindreau (2009), the book whose naming of **t** the paper follows (doi:10.1002/9780470612033
- Ettrich et al. (2014), the tensor law applied to heat conduction; only if a second reference for the law is wanted.
- A source for packing and percolation thresholds of discs, if Section 5.3 states them.

Two keys are still in the old style in the `.bib` itself: `choukroun_strength_2020` and `nicoli_tensorial_2011`.

Corrections that recur throughout

Paragraph-specific corrections are under each paragraph. These apply everywhere:

- **Notation.** Subscripts "ice" and "air" in place of i and a (ρ_{ice}, ϕ_{ice}, ϕ_{air}); $\mathbf{K}_{\text{eff}}$ for the tensor and k_{eff} for its scalar mean; k_{ice}, k_{air} as in the tables (the text has K_{ice}); ρ_{air}, P_{atm} in Eq. (2); ϕ against ϕ_i for the same field in Section 2.3.
- **Temperature.** Celsius with $T + T_0$ in Eqs. (3), (5); kelvin in Eq. (7) and Table 1. One convention.
- ε is the interface parameter, not the interface width (L208, L287).
- **"effect"** for "affect" (six places); "preform" for "perform" (three); doubled words where `\cref` already prints the word ("table table", "figure figure", "section section").
- α_c written 1.002×10^{-3}, not "1.002e-03".
- **Not cited in the text:** Table 2, Table 3, Figure 2.